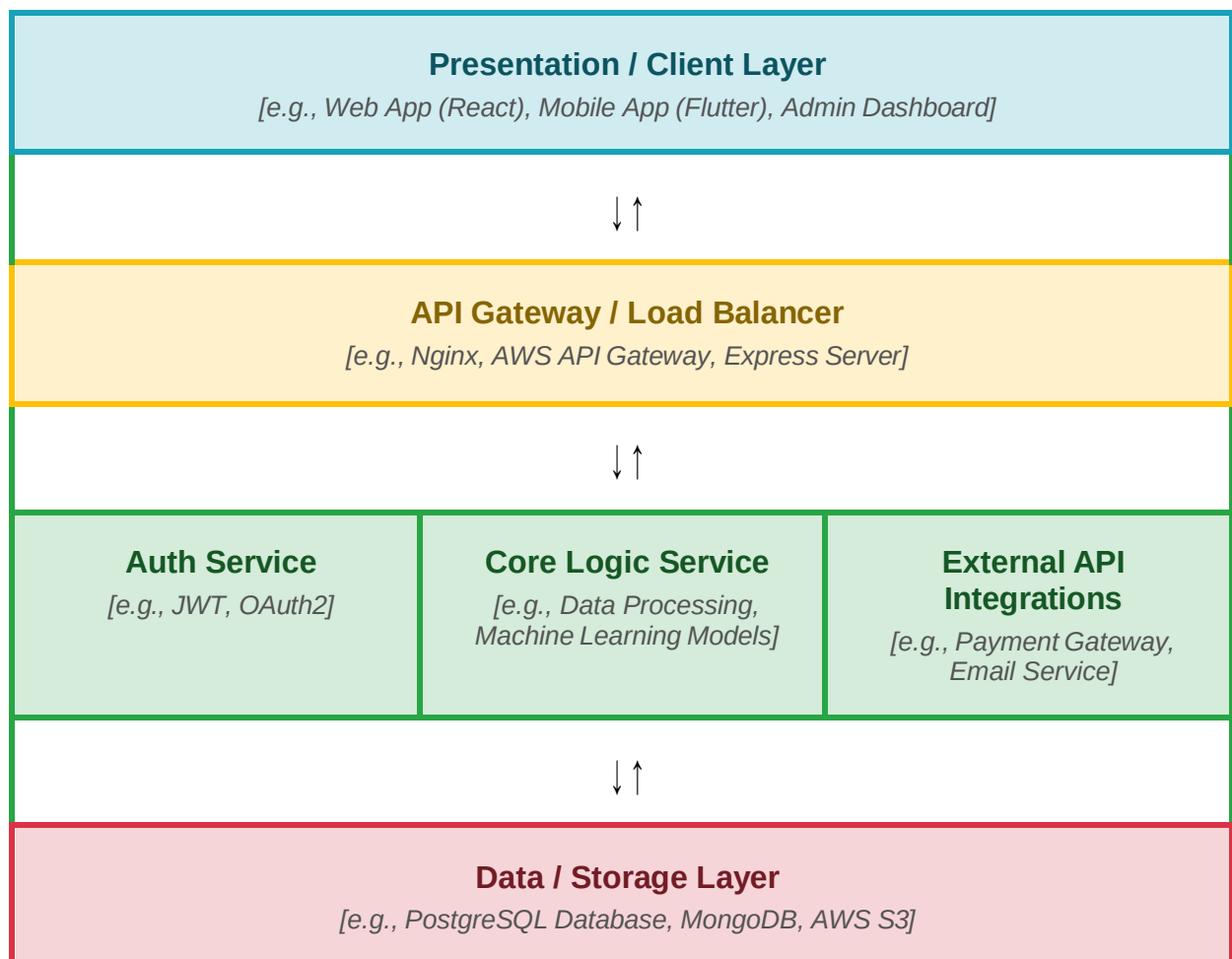


Solution Architecture

Date	30 SEPTEMBER 2026
Team ID	SWTID-2026-1418
Project Name	FIT BUDDY AI
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer (Client Layer)</i>	<i>Handles the user interface and user experience. Allows users to enter information, view workout plans, track progress, and receive reminders.</i>	<i>Flutter / Android App</i>
<i>API Gateway / Load Balancer</i>	<i>Routes requests between the mobile app and backend services. Handles API requests and communication with the server.</i>	<i>Flask REST API / Nginx</i>
<i>Core Services</i>	<i>Handles user authentication, personalized workout plans, workout recommendations, progress tracking, fitness goals, reminders, and habit support.</i>	<i>Python / Flask, JWT</i>
<i>Data / Storage Layer</i>	<i>Stores user profiles, workout details, fitness goals, progress data, activity history, and other application data.</i>	<i>MySQL / Firebase / SQLite</i>